



NORTH SYDNEY GIRLS HIGH SCHOOL
HSC Extension 1 Mathematics Assessment Task 1
Term 4, 2011

Name: _____ **Mathematics Class:** _____

Time Allowed: 60 minutes + 2 minutes reading time

Total Marks: 46

Instructions:

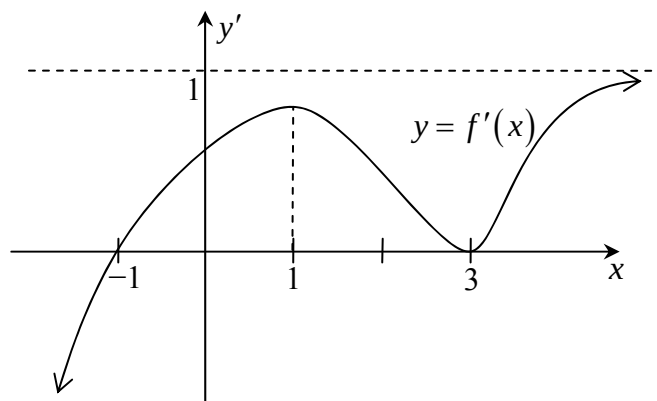
- Attempt all three questions.
- Start each question in a new booklet. Put your name on every booklet.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Work down the page.
- Do not work in columns.
- Each question will be collected separately. Submit a blank booklet if you do not attempt a question.

Question	A1-2	A3-8	B1	B2	C	Total
PE3		/14	/3			/17
HE7	/2			/11	/16	/29
						/46

Question 1 (16 marks)

(a) Write A, B, C or D on your answer sheet.

For parts (i) and (ii), consider the gradient function $y = f'(x)$ which is shown in the sketch below.



(i) The stationary points on the graph of $y = f(x)$ consist of: **1**

- (A) One local minimum and one horizontal point of inflexion.
- (B) One local maximum and one horizontal point of inflexion.
- (C) One local minimum and one local maximum.
- (D) Two horizontal points of inflexion.

(ii) The graph of $y = f(x)$ has one asymptote. **1**
The equation of this asymptote could be:

- (A) $x = 1$ (B) $y = 1$
- (C) $y = x + 2$ (D) None of the above.

(iii) The polynomial $x^3 - 2x^2 - 5x + 6$ factorises to: **1**

- (A) $(x+1)(x-2)(x+3)$ (B) $(x-1)(x+2)(x+3)$
- (C) $(x+1)(x-2)(x-3)$ (D) $(x-1)(x+2)(x-3)$

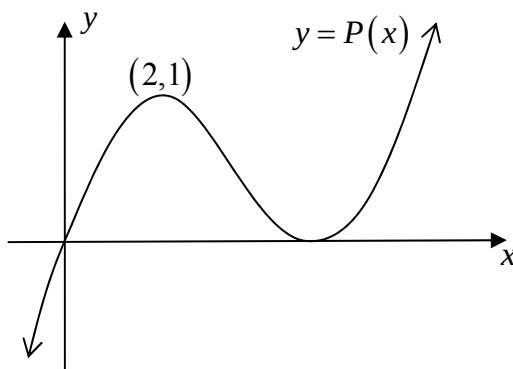
(iv) If $(x-2)^3(x+1)^2$, $(x+3)^4(x+1)$ and $(x-2)^5(x+3)^3$ are all factors of **1**
a polynomial $P(x)$, then the smallest possible degree of $P(x)$ is

- (A) 5 (B) 8
- (C) 11 (D) 18

Question 1 continued

(v)

1



The diagram shows the graph of the polynomial $y = P(x)$.

You are given that $P'(0) = 2$.

For what values of k will the equation $P(x) = kx$ have exactly three distinct roots?

(A) $k > 0$

(B) $0 < k < 2$

(C) $0 < k < 1$

(D) all real k

For the remainder of this section, show full working.

(b) (i) Show that $x - 2$ is a factor of $x^3 - 4x^2 + 3x + 2$.

1

(ii) Sketch the graph of $P(x) = x^3 - 4x^2 + 3x + 2$.

3

(iii) Hence solve the inequation $x^3 - 4x^2 + 3x + 2 < 0$.

1

(c) The polynomial $P(x)$ has a remainder of $ax - 2$ when divided by $x^2 - x - 6$,
and a remainder of 13 when divided by $x - 3$.

3

By first finding the value of a , find the remainder when $P(x)$ is divided by $x + 2$.

(d) If $5x^3 + 3x^2 - 7x - 3 = ax(x - 1)(x + 2) + bx(x - 1) + cx - 3$ for all values of x , find
the values of a, b and c .

3

Question 2 (14 marks) Start a new booklet

- (a) The polynomial equation $x^3 + bx^2 + cx + d = 0$ has roots α , β and γ . **3**

Find a simplified expression for $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$ in terms of b , c and d .

- (b) Consider the function $f(x) = \frac{x^2}{x+2}$.

- (i) Write down the equation of the vertical asymptote. **1**

- (ii) Show that $f(x) = x - 2 + \frac{4}{x+2}$. **1**

- (iii) Hence write down the equation of the non-vertical asymptote. **1**

- (iv) Show that $f'(x) = \frac{x(x+4)}{(x+2)^2}$. **3**

- (v) Find the coordinates of each stationary point and determine their nature in each case. **3**

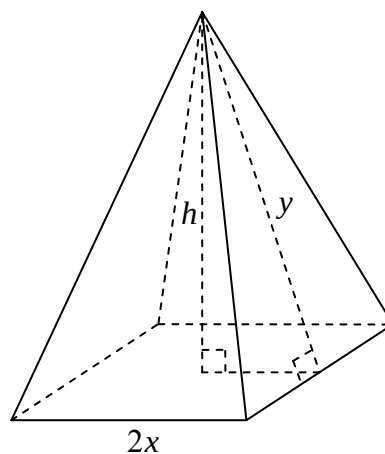
- (vi) Draw a sketch of $y = f(x)$ showing all this information. **2**

Question 3 (16 marks) Start a new booklet

(a) (i) Show that $\frac{1}{(x+h)^2} - \frac{1}{x^2} = -\frac{h(2x+h)}{x^2(x+h)^2}$. 2

(ii) Hence differentiate the function $f(x) = \frac{1}{x^2}$ by first principles. 2

- (b) A closed right pyramid has a square base of side length $2x$ cm, and its triangular faces each have perpendicular height y cm, as shown in the following diagram.



(i) This pyramid is to be constructed using 400 cm^2 of cardboard. 2
Show that $y = \frac{100}{x} - x$.

(ii) Show that the perpendicular height h of the pyramid is given by 2
$$h^2 = \frac{10000}{x^2} - 200.$$

(iii) Show that the volume V of the pyramid is given by. 2
$$V = \frac{40}{3} x \sqrt{100 - 2x^2}.$$

(iv) Show that $\frac{dV}{dx} = \frac{160(25 - x^2)}{3\sqrt{100 - 2x^2}}$. 3

(v) Find the base length and perpendicular height of such a pyramid with maximum volume. Justify your answer. 3

End of paper

Extension 1 Assessment 1 Solutions

Question 1

(a) (i) A (ii) C (iii) D (iv) C (v) B

(b) (i) Let $P(x) = x^3 - 4x^2 + 3x + 2$.

$$P(2) = 2^3 - 4(2)^2 + 3(2) + 2$$

$$= 0$$

$\therefore x - 2$ is a factor of $P(x)$

(ii) Let the zeros of $P(x)$ be α , β and 2.

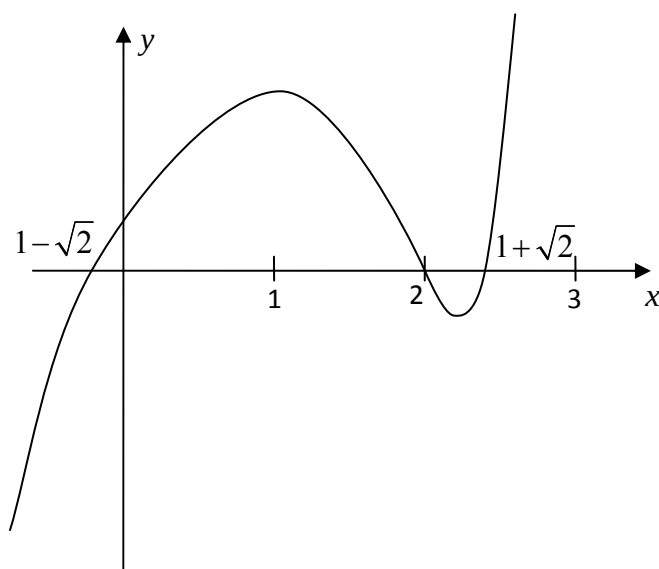
$$\text{So } \alpha + \beta + 2 = 4 \qquad 2\alpha\beta = -2$$

$$\alpha + \beta = 2 \qquad \alpha\beta = -1$$

So the quadratic with roots α and β is $x^2 - 2x - 1 = 0$

$$x = \frac{2 \pm \sqrt{8}}{2}$$

$$x = 1 \pm \sqrt{2}$$



(iii) $x < 1 - \sqrt{2}$, $2 < x < 1 + \sqrt{2}$

(c) $P(x) = (x - 3)(x + 2)Q(x) + ax - 2$

$$P(3) = 13$$

$$13 = 3a - 2$$

$$a = 5$$

$$\text{remainder} = P(-2)$$

$$= -12$$

(d) $5x^3 + 3x^2 - 7x - 3 = ax(x - 1)(x + 2) + bx(x - 1) + cx - 3$

$$(x = 1): \quad 5 + 3 - 7 - 3 = c - 3 \quad \Rightarrow \quad c = 1$$

$$(x = -2): \quad -40 + 12 + 14 - 3 = 6b - 2 - 3 \quad \Rightarrow \quad b = -2$$

$$(x = 2): \quad 40 + 12 - 14 - 3 = 8a - 4 + 2 - 3 \quad \Rightarrow \quad a = 5$$

Question 2

$$\begin{aligned}
 \text{(a)} \quad \frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta} &= \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} \\
 &= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)}{\alpha\beta\gamma} \\
 &= \frac{(-b)^2 - 2(c)}{-d} \\
 &= \frac{2c - b^2}{d}
 \end{aligned}$$

(b) (i) $x = -2$

$$\begin{aligned}
 \text{(ii)} \quad RHS &= x - 2 + \frac{4}{x+2} \\
 &= \frac{(x-2)(x+2) + 4}{x+2} \\
 &= \frac{x^2 - 4 + 4}{x+2} \\
 &= \frac{x^2}{x+2}
 \end{aligned}$$

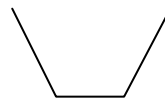
(iii) $y = x - 2$

(iv)

$$\begin{aligned}
 f'(x) &= \frac{(x+2) \cdot 2x - x^2 \cdot 1}{(x+2)^2} \\
 &= \frac{2x^2 + 4x - x^2}{(x+2)^2} \\
 &= \frac{x^2 + 4x}{(x+2)^2} \\
 &= \frac{x(x+4)}{(x+2)^2}
 \end{aligned}$$

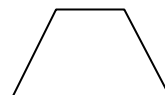
(v) Stationary points at $x = 0, -4$.

x	-1	0	1
$\frac{dy}{dx}$	-3	0	$\frac{5}{9}$



so, local minimum at $(0, 0)$

x	-5	-4	-3
$\frac{dy}{dx}$	$\frac{5}{9}$	0	-3



so, local maximum at $(-4, -8)$

(v) [Alternative solution]

$$f(x) = x - 2 + \frac{4}{x+2}$$
$$= x - 2 + 4(x+2)^{-1}$$

$$f'(x) = 1 - 4(x+2)^{-2}$$
$$= 1 - \frac{4}{(x+2)^2}$$

$$f''(x) = 8(x+2)^{-3}$$
$$= \frac{8}{(x+2)^3}$$

Stat Pts: $1 - \frac{4}{(x+2)^2} = 0$

$$(x+2)^2 = 4$$

$$x = 0, -4$$

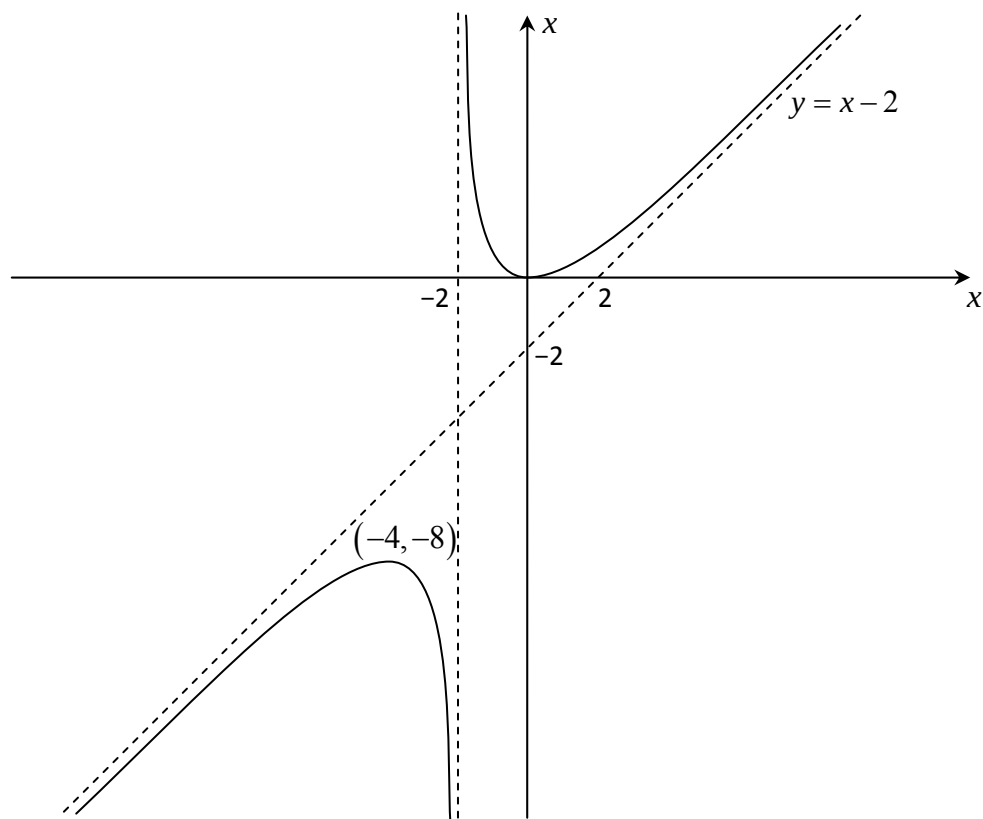
$$f(0) = 0, f''(0) = 1 > 0$$

so, local min. at $(0, 0)$

$$f(-4) = -8, f''(-4) = -1 < 0$$

so, local max. at $(-4, -8)$

(vi)



Question 3

(a) (i)

$$\begin{aligned}\frac{1}{(x+h)^2} - \frac{1}{x^2} &= \frac{x^2 - (x+h)^2}{x^2(x+h)^2} \\&= \frac{x^2 - x^2 - 2xh - h^2}{x^2(x+h)^2} \\&= \frac{-2xh - h^2}{x^2(x+h)^2} \\&= -\frac{h(2x+h)}{x^2(x+h)^2}\end{aligned}$$

(ii)

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} \\&= \lim_{h \rightarrow 0} \frac{-h(2x+h)}{hx^2(x+h)^2} \\&= \frac{-2x}{x^2 \cdot x^2} \\&= -\frac{2}{x^3}\end{aligned}$$

(b) (i) $4x^2 + 4\left(\frac{1}{2} \cdot 2x \cdot y\right) = 400$

$$x^2 + xy = 100$$

$$xy = 100 - x^2$$

$$y = \frac{100}{x} - x$$

(ii) $h^2 = y^2 - x^2$ by Pythagoras' Theorem

$$= \left(\frac{100}{x} - x\right)^2 - x^2$$

$$= \frac{10000}{x^2} - 200 + x^2 - x^2$$

$$= \frac{10000}{x^2} - 200$$

(iii)

$$\begin{aligned} V &= \frac{1}{3}(2x)^2 h \\ &= \frac{4}{3}x^2 \sqrt{\frac{10000}{x^2} - 200} \\ &= \frac{4}{3}x^2 \cdot \frac{10}{x} \sqrt{100 - 2x^2} \\ &= \frac{40}{3}x \sqrt{100 - 2x^2} \end{aligned}$$

(iv)

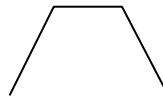
$$\begin{aligned} \frac{dV}{dx} &= \frac{40}{3} \left[(100 - 2x^2)^{\frac{1}{2}} \cdot 1 + x \cdot \frac{1}{2} (100 - 2x^2)^{-\frac{1}{2}} \cdot (-4x) \right] \\ &= \frac{40}{3} \left[(100 - 2x^2)^{\frac{1}{2}} - 2x^2 (100 - 2x^2)^{-\frac{1}{2}} \right] \\ &= \frac{40}{3} (100 - 2x^2)^{-\frac{1}{2}} [(100 - 2x^2) - 2x^2] \\ &= \frac{160}{3} (100 - 2x^2)^{-\frac{1}{2}} (25 - x^2) \\ &= \frac{160(25 - x^2)}{3\sqrt{100 - 2x^2}} \end{aligned}$$

(v)

$$\frac{dV}{dx} = 0$$

$$x = 5 \quad (x > 0)$$

x	4	5	6
$\frac{dV}{dx}$	$\frac{240}{\sqrt{17}}$	0	$\frac{-880}{3\sqrt{7}}$



i.e. maximum Volume when $x = 5$

Dimensions:

Base: 10 cm

Perp Ht: $10\sqrt{2}$ cm